

# FreshFlow Beverages Pvt. Ltd.

FreshFlow Bottling Plant - Pune Production Facility  
FSSAI: 10012022000345 | ISO 22000:2018 / ISO 14001:2015

## EQUIPMENT MANUAL

### Tetra Pak TK103 Water Storage Tank - Operation and Maintenance Manual

Model: FF-TK-176 | Serial: FFB-TK-26701

**Equipment ID: TK103**

|                                                  |
|--------------------------------------------------|
| Document Number: MAN-TK103-001                   |
| Revision: Rev 2.1                                |
| Classification: CONFIDENTIAL - INTERNAL USE ONLY |
| Prepared by: Engineering / QA Department         |
| Approved by: Plant Manager                       |

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## 1. Introduction and Equipment Description

The TK103 Water Storage Tank (Asset ID: TK103) is a SS316 Water Storage Tank manufactured by Tetra Pak, model FF-TK-176, serial number FFB-TK-26701. It is installed in the Water Treatment department at FreshFlow Bottling Plant - Pune Production Facility. The equipment was commissioned on 2020-05-15 with a warranty valid until N/A.

The TK103 Water Storage Tank is classified as High criticality equipment with an expected operational life of 15 years. It requires monthly preventive maintenance as per the FreshFlow Beverages Asset Management Program (AMP). The equipment is located at: Water Treatment Zone 1.

The equipment operates at a rated speed of 1450 RPM with a power input of 11 kW at full load.

All maintenance and operational activities must comply with FreshFlow Beverages Pvt. Ltd. safety and food hygiene standards, ISO 22000:2018, and FSSAI regulations. Only trained and authorized personnel may operate or work on this equipment.

### 1.1 Purpose of This Manual

This manual provides complete guidance for safe operation, preventive maintenance, troubleshooting, and emergency procedures for the above equipment. All personnel working on or near this equipment must read and understand this manual before starting work.

## 2. Technical Specifications

| Parameter        | Value       |
|------------------|-------------|
| Storage Capacity | 50 KL       |
| Material         | SS316L      |
| Design Pressure  | Atmospheric |

### 2.1 Attached Sensors and Instruments

| Sensor / Parameter  | Monitoring Mode       | Data System       |
|---------------------|-----------------------|-------------------|
| Tank Level          | Continuous monitoring | SCADA / Historian |
| Water Quality       | Continuous monitoring | SCADA / Historian |
| Machine Temperature | Continuous monitoring | SCADA / Historian |

### 2.2 Design Parameters and Limits

| Design Parameter                          | Value                  |
|-------------------------------------------|------------------------|
| Minimum Operating Temperature             | -1°C                   |
| Maximum Operating Temperature             | 121°C                  |
| Design Pressure                           | 11 bar(g)              |
| Maximum Allowable Working Pressure (MAWP) | 14 bar(g)              |
| Noise Level (at 1m)                       | 82 dB(A)               |
| IP Protection Class                       | IP65                   |
| Hazardous Area Classification             | Zone 2 (if applicable) |

### 3. Safety Instructions and Warnings

#### ! DANGER

DANGER: This equipment contains rotating parts, high pressure, and electrical hazards. Failure to follow safety instructions may result in severe injury or death. ALWAYS apply Lockout/Tagout (LOTO) before any maintenance work. See SOP-SAF-002.

#### ! WARNING

FOOD SAFETY: This equipment contacts food-grade product. Only food-grade lubricants, gaskets, and cleaning chemicals approved by FreshFlow QA department may be used. Unauthorized materials may cause product contamination.

#### 3.1 Mandatory PPE

- Safety helmet (bump cap within the production area)
- Safety shoes (steel-toed, anti-slip, food grade)
- Safety glasses / goggles (when working with chemicals or pressurised systems)
- Cut-resistant gloves (when handling sharp parts - NOT when working near rotating machinery)
- Hearing protection ( $\geq 85$  dB areas)
- High-visibility vest when working in forklift zone

#### 3.2 Lockout / Tagout (LOTO)

1. Notify the shift supervisor and control room before starting any maintenance.
2. Isolate all energy sources: electrical (main isolator), pneumatic (air supply valve), steam (isolation valve), and hydraulic as applicable.
3. Apply personal padlock and danger tag on each isolation point.
4. Verify zero energy state: test electrical with approved voltage tester, release trapped pressure.
5. Perform maintenance work.
6. Remove all tools, materials and personnel from equipment before re-energizing.
7. Remove LOTO in reverse order. Notify operator before restart.

#### ! CAUTION

Never bypass safety interlocks, guards, or alarms for any reason. Bypassing interlocks requires a formal bypass authorization form signed by the Plant Manager.

## 4. Installation and Commissioning

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### 4.1 Foundation and Leveling

Ensure the concrete foundation is fully cured (minimum 28 days) before placing equipment. Level the baseplate with stainless steel shims. Maximum allowable deviation is 0.05 mm/m. Perform a foundation bolt check to ensure all anchor bolts are correctly embedded and tensioned to the design torque specified in the equipment drawing.

### 4.2 Piping and Connection Requirements

All piping connected to this equipment must be independently supported to avoid imposing mechanical loads on the equipment nozzles. Maximum allowable nozzle loads must not exceed values specified by the manufacturer. Use food-grade gaskets (EPDM or PTFE) at all flanged joints. Ensure all pipe work is clean and free from debris before connection.

### 4.3 Electrical Connection

Electrical connections must be performed by a certified electrician in accordance with IS:732, the Indian Electricity Rules, and FreshFlow Beverages Electrical Safety Standards. Verify voltage, frequency, and phase sequence before energizing. Confirm protective earth is correctly connected. Record installation details in the Equipment History Register.

### 4.4 Pre-Commissioning Checks

1. Verify all mechanical fasteners are tightened to specified torques.
2. Check direction of rotation before full coupling - bump test only.
3. Fill lubricant/oil to correct level per sight glass. Use OEM-specified grade only.
4. Open all isolation valves to the correct position per P&ID.
5. Verify all instrument connections (pressure gauges, temperature sensors, flow meters).
6. Test all alarms and interlocks with SCADA team before commissioning at load.
7. Complete and sign the Pre-Commissioning Checklist (Form FFB-ENG-PC-01).

## 5. Startup Procedure

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### 5.1 Pre-Start Checks

- Confirm LOTO has been removed and all tools are cleared from equipment.
- Verify no active alarms on TK103 on SCADA / local panel.
- Confirm lubricant and cooling fluid levels are within operating range.
- Confirm Water Treatment department is ready to receive flow/product.
- Check all guards are fitted securely. Do not start with any guard removed.
- Verify compressed air supply pressure is 6-7 bar (for pneumatic components).

### 5.2 Startup Sequence

1. Acknowledge all start permissives on SCADA for equipment TK103.
2. Start cooling / utility systems first if applicable (chilled water, steam, air).
3. Engage the main driver at low speed (if VFD) or direct start as per control configuration.
4. Gradually bring equipment to operating setpoint. Monitor all parameters.
5. Verify operating parameters are within normal range within 5 minutes of start.
6. Log startup time in shift log and notify the Water Treatment supervisor.

#### ! WARNING

If any alarm activates during startup, stop the equipment immediately. Investigate the cause before attempting a second start. Do not repeatedly attempt to force start without investigating.

## 6. Normal Operating Procedure

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Under normal operating conditions, TK103 Water Storage Tank should operate continuously without operator intervention. The SCADA system monitors all critical parameters and will alarm if any value exceeds the set limits. The operator must:

- Perform hourly visual checks of the equipment during operation.
- Verify the running parameters against the normal operating range in Section 2.
- Record any unusual observations in the shift log immediately.
- Report any abnormal noise, vibration, leakage or temperature to the maintenance team.
- Never attempt to adjust process parameters beyond the approved setpoints without authorization.

## 7. Shutdown Procedure

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### 7.1 Planned Shutdown

1. Notify the Water Treatment supervisor and control room of the planned shutdown.
2. Reduce equipment to minimum load / speed before stopping.
3. Initiate stop command from SCADA or local panel as appropriate.
4. Allow equipment to coast to rest naturally - do not apply manual braking.
5. Close isolation valves (suction/discharge/utility) as per P&ID.
6. For food contact equipment: Initiate CIP sequence if required before shutdown.
7. Apply LOTO if the shutdown is for maintenance purpose.
8. Log shutdown time, reason, and duration in the equipment history record.

### 7.2 Emergency Shutdown

#### **! DANGER**

Emergency stop buttons are located on the local control panel and at the filling area main panel. Press the red E-STOP button only in case of emergency. E-STOP activates the electrical isolation of all drives simultaneously.

## 8. Preventive Maintenance Schedule

| Task Description                                    | Frequency | Responsible            | Record            |
|-----------------------------------------------------|-----------|------------------------|-------------------|
| Visual inspection - leaks, damage, loose fasteners  | Daily     | Line Operator          | Operator Log      |
| Check lubricant levels and colour                   | Daily     | Line Operator          | PM Checklist      |
| Vibration and temperature monitoring                | Weekly    | Reliability Technician | Historian         |
| Clean exterior and ventilation / heat exchange area | Monthly   | Maintenance            | PM Work Order     |
| Lubricate bearings/chain per OEM schedule           | Monthly   | Maintenance            | PM Work Order     |
| Inspect seals, gaskets and O-rings for wear         | Monthly   | Maintenance            | PM Work Order     |
| Check alignment and coupling condition              | Quarterly | Maintenance            | PM Work Order     |
| Filter element inspection/replacement               | Quarterly | Maintenance            | PM Work Order     |
| Oil analysis / oil change                           | 6 Months  | Reliability Tech       | Oil Sample Report |
| Foundation bolt torque verification                 | Yearly    | Maintenance            | Torque Record     |
| Comprehensive overhaul / OEM inspection             | 3 Years   | OEM Specialist         | Overhaul Report   |

All preventive maintenance must be performed under Lockout/Tagout (SOP-SAF-002) and recorded in the FreshFlow CMMS system with work order number, technician ID, and parts used. Any deviations from the above schedule require Plant Manager approval.



## 9. Troubleshooting Guide

| Symptom                    | Probable Cause                         | Corrective Action                                                     |
|----------------------------|----------------------------------------|-----------------------------------------------------------------------|
| Performance drop           | Fouling or wear                        | Clean or overhaul equipment as per scheduled maintenance              |
| Unusual noise/vibration    | Mechanical looseness or bearing issue  | Inspect, tighten fasteners and replace bearings as required           |
| Overheating                | Lack of lubrication or cooling failure | Check lube levels/coolant. Verify cooling flow                        |
| Control/sensor failure     | Sensor drift or wiring fault           | Calibrate or replace sensor. Check wiring continuity                  |
| Leakage (product or fluid) | Seal or gasket degradation             | Replace gasket/seal. Use food-grade material per plant specifications |

If the above corrective actions do not resolve the problem, escalate to the Reliability Engineer (EMP016 - Ajay Mane) or contact the OEM technical support. Record all troubleshooting steps in the CMMS maintenance log.

## 10. CIP and Sanitation Requirements

All equipment that contacts food product must undergo Clean-In-Place (CIP) at the defined frequency. CIP chemicals, concentrations, temperatures, and contact times are defined in the FreshFlow CIP Master Plan (FFB-QA-CIP-001).

### 10.1 Standard CIP Parameters

| Parameter                     | Value                                             |
|-------------------------------|---------------------------------------------------|
| Pre-rinse with water          | 5 min   25°C   until runoff is clear              |
| Caustic wash (NaOH)           | 20 min   75°C   2% NaOH (food grade)              |
| Intermediate water rinse      | 5 min   25°C   until pH neutral                   |
| Acid rinse (HNO3)             | 15 min   65°C   1% HNO3 (passivation)             |
| Final water rinse             | 10 min   25°C   RO water (conductivity <20 uS/cm) |
| Sanitisation (Peracetic Acid) | 10 min   25°C   150 ppm PAA (if required)         |

**! NOTE**

CIP verification is mandatory after every CIP cycle. The final rinse water must test <=20 uS/cm conductivity AND pH 6.5-7.5 AND a microbiological swab must confirm <10 CFU/cm2 before the equipment is released for production.

## 11. Spare Parts Catalog

| Part Description                 | Quantity (Min Stock) | OEM Notes                      |
|----------------------------------|----------------------|--------------------------------|
| Mechanical Seal / Seal Cartridge | 1 set                | Specific to FF-TK-176          |
| Bearing (Drive End)              | 1 no.                | Per OEM specification          |
| Bearing (Non-Drive End)          | 1 no.                | Per OEM specification          |
| Coupling Spider / Insert         | 1 no.                | Per coupling model             |
| Gasket Set                       | 1 set                | Full set per overhaul schedule |
| O-Ring Kit                       | 1 set                | Food-grade material only       |
| Oil Filter Element               | 2 nos.               | Per OEM part number            |
| Air Filter Element               | 2 nos.               | Per OEM part number            |
| Pressure Gauge (0-10 bar)        | 1 no.                | Class 1.0 accuracy             |
| Thermocouple / RTD               | 1 no.                | Match to existing sensor type  |

All spare parts must be sourced from authorized suppliers and must be food-grade compatible where applicable. Store spares in clean, dry conditions in the FreshFlow Spare Parts Store (Block K). Minimum stock levels must be maintained at all times.

## 12. Alarms, Interlocks and Set Points

| Alarm / Interlock            | Set Point                     | Required Action                               |
|------------------------------|-------------------------------|-----------------------------------------------|
| Temperature High (Warning)   | 85°C                          | SCADA alarm - investigate                     |
| Temperature High-High (Trip) | 98°C                          | Auto-trip - investigate before restart        |
| Pressure High                | 8 bar                         | SCADA alarm - check relief valve              |
| Vibration High (Warning)     | 7 mm/s                        | Investigate bearing/alignment                 |
| Vibration High-High (Trip)   | 12 mm/s                       | Auto-trip - do not restart until investigated |
| Motor Current High           | 110% FLC                      | SCADA alarm - check mechanical load           |
| Low Pressure                 | 1 bar                         | Check supply / suction conditions             |
| Leak Detected                | Float switch / flow deviation | Stop equipment - investigate immediately      |

## 13. Emergency Procedures

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### 13.1 Product Spill / Contamination

In the event of a product leak or contamination event: Stop the filling/transfer immediately. Isolate and hold all product affected. Notify QC Manager (EMP003). Do not release any product until QC clearance is obtained. Record in incident log.

### 13.2 Chemical Spill (Caustic / Acid)

In case of CIP chemical spill: Alert personnel in the area immediately. Evacuate if large volume (>10 litres). Do not enter without chemical-resistant PPE (nitrile gloves, face shield, apron). Neutralize caustic with dilute acid and vice versa. Wash area with large volumes of water. Notify HSE Officer (EMP009 / EMP017).

### 13.3 CO2 Gas Leak (if applicable)

If CO2 detector GD101 alarms: Evacuate the CO2 zone immediately. Do not re-enter until GD101 reading below 0.5% and confirmed by HSE Officer. CO2 is heavier than air and accumulates at floor level - it causes asphyxiation. Activate exhaust fan EX101 remotely from Control Room. See SOP-SAF-001.

## 14. Appendix - Reference Documents

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- P&ID Drawing: FFB-ENG-PID-TK103-001 Rev 3
- Electrical Drawing: FFB-ENG-EL-TK103-001 Rev 2
- Foundation Drawing: FFB-ENG-CV-TK103-001 Rev 1
- OEM Manual: Tetra Pak FF-TK-176 Original OEM Manual
- CMMS Tag: TK103 - FreshFlow Beverages CMMS
- Risk Assessment: FFB-HSE-RA-TK103-001
- Calibration Record: FFB-QA-CAL-TK103-001
- CIP Validation: FFB-QA-CIP-TK103-001

For the most current version of this manual and related documents, access the FreshFlow Beverages Document Management System (DMS) at the Control Room terminal or contact the Engineering Department.

## A. Energy Efficiency Notes

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The TK103 Water Storage Tank consumes approximately 11.0 kW at full load. Energy efficiency improvements can be achieved through VFD speed optimization, correct operating point selection, and regular maintenance to minimize friction losses. Track kWh/hour in the SCADA historian and compare with baseline established at commissioning.

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## A. Food Safety Considerations

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All equipment in product contact zones must be maintained to prevent contamination risks. This includes: using only food-grade lubricants (NSF H1 certified), ensuring all seals and gaskets are food-grade materials (EPDM, PTFE or silicone), maintaining equipment hygiene zones as per the FreshFlow Hygienic Equipment Standard (HES-001), and reporting any metallic part damage immediately to QC to trigger a metal contamination investigation protocol.

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## A. Environmental and Sustainability Notes

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FreshFlow Beverages is committed to environmental sustainability. All maintenance activities must minimize waste generation. Used oil, chemicals and filters must be disposed of through the approved hazardous waste contractor (GreenEco Pvt. Ltd.) per the Waste Disposal Plan (FFB-HSE-WDP-001). Water usage during CIP must be tracked and reported monthly.

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## A. Revision History

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- Rev 1.0 (2021-10-01): Initial release at plant commissioning.
- Rev 1.5 (2022-06-01): Updated troubleshooting section after first year operational review.
- Rev 2.0 (2023-07-01): Major update - added CIP parameters and food safety section.
- Rev 2.1 (2024-01-01): Alarm setpoints updated per Reliability Engineering recommendation.
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- Rev 2.1 (2024-01-01): Alarm setpoints updated per Reliability Engineering recommendation.

## A. Technical Notes - Vibration Analysis

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This section provides additional technical guidance for maintenance engineers. Vibration analysis using route-based data collection (monthly) and online monitoring (continuous) allows early detection of bearing defects, misalignment, imbalance, and looseness. ISO 10816-3 provides the vibration acceptance criteria for industrial machinery. For TK103 Water Storage Tank, the acceptable vibration limit is 4.5 mm/s RMS at the bearing housing. Any reading above 7.1 mm/s requires immediate investigation and remedial action. Lubrication selection must follow the OEM specification. For rolling element bearings, use lithium complex grease NLGI Grade 2 with food-grade additive package (NSF H1). The regreasing interval is calculated based on the Lubrication Engineer's formula incorporating speed, bearing size, and operating temperature.

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## A. Technical Notes - Lubrication Engineering

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This section provides additional technical guidance for maintenance engineers. Vibration analysis using route-based data collection (monthly) and online monitoring (continuous) allows early detection of bearing defects, misalignment, imbalance, and looseness. ISO 10816-3 provides the vibration acceptance criteria for industrial machinery. For TK103 Water Storage Tank, the acceptable vibration limit is 4.5 mm/s RMS at the bearing housing. Any reading above 7.1 mm/s requires immediate investigation and remedial action. Lubrication selection must follow the OEM specification. For rolling element bearings, use lithium complex grease NLGI Grade 2 with food-grade additive package (NSF H1). The regreasing interval is calculated based on the Lubrication Engineer's formula incorporating speed, bearing size, and operating temperature.

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## A. Technical Notes - Alignment Standards

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